

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Integrals

1.1 Medium Integral

Inverse Hyperbolic Tangent Integral

Evaluate the definite integral:

$$\int_0^{\pi/3} \sec^2(x) \tanh^{-1}(\sin(x)) dx$$

Solution: Let the target integral be denoted by I . The presence of $\sec^2(x)$ alongside a composite function of $\sin(x)$ makes integration by parts the ideal technique. We choose our parts as follows:

$$u = \tanh^{-1}(\sin(x))$$

$$dv = \sec^2(x) dx$$

Differentiating u with respect to x involves the chain rule. Recall that $\frac{d}{dz} \tanh^{-1}(z) = \frac{1}{1-z^2}$:

$$du = \frac{1}{1-\sin^2(x)} \cdot \cos(x) dx = \frac{\cos(x)}{\cos^2(x)} dx = \sec(x) dx$$

Integrating dv yields:

$$v = \tan(x)$$

Applying the integration by parts formula $\int u dv = uv - \int v du$:

$$I = \left[\tan(x) \tanh^{-1}(\sin(x)) \right]_0^{\pi/3} - \int_0^{\pi/3} \tan(x) \sec(x) dx$$

The remaining integral is a standard elementary form, as $\frac{d}{dx} \sec(x) = \sec(x) \tan(x)$:

$$I = \left[\tan(x) \tanh^{-1}(\sin(x)) - \sec(x) \right]_0^{\pi/3}$$

Now, we evaluate the expression at the upper and lower limits. At the upper limit $x = \pi/3$:

$$\tan(\pi/3) = \sqrt{3}, \quad \sin(\pi/3) = \frac{\sqrt{3}}{2}, \quad \sec(\pi/3) = 2$$

$$\tanh^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{1}{2} \ln\left(\frac{1+\frac{\sqrt{3}}{2}}{1-\frac{\sqrt{3}}{2}}\right) = \frac{1}{2} \ln\left(\frac{2+\sqrt{3}}{2-\sqrt{3}}\right)$$

Rationalizing the argument of the logarithm by multiplying numerator and denominator by $2 + \sqrt{3}$:

$$\tanh^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{1}{2} \ln\left(\frac{(2+\sqrt{3})^2}{4-3}\right) = \ln(2+\sqrt{3})$$

So, the upper limit evaluation yields: $\sqrt{3}\ln(2 + \sqrt{3}) - 2$. At the lower limit $x = 0$:

$$\tan(0) = 0, \quad \sec(0) = 1$$

The lower limit evaluation yields: $0 - 1 = -1$. Subtracting the lower limit value from the upper limit value gives the exact result:

$$I = (\sqrt{3}\ln(2 + \sqrt{3}) - 2) - (-1)$$

$$I = \sqrt{3}\ln(2 + \sqrt{3}) - 1$$

1.2 Hard Integral

Logarithmic Rational Integral

Evaluate the definite integral:

$$\int_0^1 \frac{\ln(1 + x + x^2)}{x(1 + x)} dx$$

Solution: To evaluate this challenging integral, we employ differentiation under the integral sign (Feynman's trick). We introduce a parameter a and define the function:

$$I(a) = \int_0^1 \frac{\ln(1 + ax + x^2)}{x(1 + x)} dx$$

We want to find $I(1)$. Differentiating $I(a)$ with respect to a :

$$I'(a) = \int_0^1 \frac{x}{x(1 + x)(1 + ax + x^2)} dx = \int_0^1 \frac{1}{(1 + x)(1 + ax + x^2)} dx$$

Using partial fraction decomposition:

$$\frac{1}{(1 + x)(1 + ax + x^2)} = \frac{A}{1 + x} + \frac{Bx + C}{1 + ax + x^2}$$

Solving for the constants gives $A = \frac{1}{2-a}$, $B = \frac{-1}{2-a}$, and $C = \frac{1-a}{2-a}$. Thus:

$$I'(a) = \frac{1}{2-a} \int_0^1 \left(\frac{1}{1+x} - \frac{x+a-1}{1+ax+x^2} \right) dx$$

The integral of the first term is simply $\ln(2)$. For the second term, we rewrite the numerator:

$$\frac{x+a-1}{1+ax+x^2} = \frac{1}{2} \frac{2x+a}{1+ax+x^2} - \frac{2-a}{2} \frac{1}{1+ax+x^2}$$

Integrating this broken-down expression from 0 to 1:

$$\int_0^1 \frac{x+a-1}{1+ax+x^2} dx = \frac{1}{2} \left[\ln(1+ax+x^2) \right]_0^1 - \frac{2-a}{2} \int_0^1 \frac{1}{(x+a/2)^2 + 1 - a^2/4} dx$$

$$= \frac{1}{2} \ln(2+a) - \frac{2-a}{\sqrt{4-a^2}} \left[\tan^{-1} \left(\frac{2x+a}{\sqrt{4-a^2}} \right) \right]_0^1$$

Substituting this back into $I'(a)$ and simplifying heavily:

$$I'(a) = \frac{\ln(2) - \frac{1}{2} \ln(2+a)}{2-a} + \frac{1}{\sqrt{4-a^2}} \left(\tan^{-1} \sqrt{\frac{2+a}{2-a}} - \tan^{-1} \frac{a}{\sqrt{4-a^2}} \right)$$

By substituting $a = 2 \sin(\alpha)$, the arctangent terms cleanly reduce to $\frac{\pi}{4} - \frac{\alpha}{2}$. To find $I(1)$, we use the Fundamental Theorem of Calculus: $I(1) = I(2) - \int_1^2 I'(a) da$. At $a = 2$, the original integral is trivial:

$$I(2) = \int_0^1 \frac{\ln(1+2x+x^2)}{x(1+x)} dx = 2 \int_0^1 \frac{\ln(1+x)}{x(1+x)} dx = 2 \left(\frac{\pi^2}{12} - \frac{1}{2} \ln^2(2) \right) = \frac{\pi^2}{6} - \ln^2(2)$$

Now, evaluating $\int_1^2 I'(a) da$ involves integrating the two constituent terms of $I'(a)$. The trigonometric term integrates directly via the $a = 2 \sin(\alpha)$ substitution yielding $\frac{\pi^2}{36}$. The logarithmic term integrates to $\frac{1}{2} \text{Li}_2(1/4)$ by substituting $u = 2 - a$. Subtracting the accumulated integral values from $I(2)$ yields our final exact closed form:

$$I(1) = \frac{\pi^2}{6} - \ln^2(2) - \frac{\pi^2}{36} - \frac{1}{2} \text{Li}_2 \left(\frac{1}{4} \right)$$

$$I(1) = \frac{5\pi^2}{36} - \ln^2(2) - \frac{1}{2} \text{Li}_2 \left(\frac{1}{4} \right)$$

Differentials

2.1 Medium Differential

First-Order Non-Linear Differential Equation

Solve the following differential equation:

$$y'y + x = \sqrt{x^2 + y^2}$$

Given the initial condition $y(0) = 1$, find the exact value of $y(1)$.

Solution: We rewrite the differential equation slightly to recognize a familiar differential form:

$$y \frac{dy}{dx} + x = \sqrt{x^2 + y^2}$$

Notice that the left-hand side is exactly half the derivative of the expression inside the square root on the right-hand side. We apply the substitution $u = x^2 + y^2$. Differentiating implicitly with respect to x yields:

$$\frac{du}{dx} = 2x + 2y \frac{dy}{dx} \implies \frac{1}{2} \frac{du}{dx} = x + y \frac{dy}{dx}$$

Substituting this into our original differential equation transforms it into a separable first-order equation in u :

$$\frac{1}{2} \frac{du}{dx} = \sqrt{u}$$

Separating variables gives:

$$\frac{du}{2\sqrt{u}} = dx$$

Integrating both sides:

$$\sqrt{u} = x + C$$

Substituting back $u = x^2 + y^2$:

$$\sqrt{x^2 + y^2} = x + C$$

We now use the initial condition $y(0) = 1$ to determine the constant of integration C :

$$\sqrt{0^2 + 1^2} = 0 + C \implies C = 1$$

The particular solution is therefore:

$$\sqrt{x^2 + y^2} = x + 1$$

To find $y(x)$, we square both sides and isolate y :

$$x^2 + y^2 = (x + 1)^2$$

$$\begin{aligned}x^2 + y^2 &= x^2 + 2x + 1 \\y^2 &= 2x + 1\end{aligned}$$

Since our initial condition $y(0) = 1$ is positive, we take the positive principal root:

$$y(x) = \sqrt{2x + 1}$$

Finally, we evaluate the function at $x = 1$:

$$y(1) = \sqrt{2(1) + 1} = \sqrt{3}$$

2.2 Hard Differential

Non-Linear ODE with Incomplete Gamma Function

Consider the differential equation:

$$y \frac{d^2 y}{dx^2} - 2 \left(\frac{dy}{dx} \right)^2 = y^3 e^{-y} \left(\frac{dy}{dx} \right)^3$$

If the general solution takes the form:

$$x(y) = \frac{1}{y} (\gamma(A, y) + c_1) - \gamma(B, y) + c_2$$

where $\gamma(n, y) = \int_0^y t^{n-1} e^{-t} dt$ is the lower incomplete gamma function, find the value of $A + B$.

Solution: The structure of the solution and the heavy presence of powers of $\frac{dy}{dx}$ hint that we should invert the roles of the variables, treating x as the dependent variable and y as the independent variable. Let $x' = \frac{dx}{dy}$. The derivatives transform as follows:

$$\frac{dy}{dx} = \frac{1}{x'}$$

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{1}{x'} \right) = \frac{d}{dy} \left(\frac{1}{x'} \right) \frac{dy}{dx} = \left(-\frac{x''}{(x')^2} \right) \frac{1}{x'} = -\frac{x''}{(x')^3}$$

Substituting these transformed derivatives into the original equation:

$$y \left(-\frac{x''}{(x')^3} \right) - 2 \left(\frac{1}{x'} \right)^2 = y^3 e^{-y} \left(\frac{1}{x'} \right)^3$$

Multiplying the entire equation by $-(x')^3$ cleanly removes the fractions:

$$yx'' + 2x' = -y^3 e^{-y}$$

Dividing by y puts the equation into standard linear first-order form for the variable $v = x'$:

$$x'' + \frac{2}{y} x' = -y^2 e^{-y} \implies v' + \frac{2}{y} v = -y^2 e^{-y}$$

The integrating factor for this linear ODE is $\mu = \exp(\int \frac{2}{y} dy) = \exp(2 \ln(y)) = y^2$. Multiplying through by the integrating factor gives:

$$y^2 v' + 2yv = -y^4 e^{-y} \implies \frac{d}{dy}(y^2 v) = -y^4 e^{-y}$$

Integrating both sides with respect to y :

$$y^2 v = \int -y^4 e^{-y} dy + C_1$$

Recall the lower incomplete gamma function definition: $\gamma(n, y) = \int_0^y t^{n-1} e^{-t} dt$. The integral $\int -y^4 e^{-y} dy$ corresponds to an antiderivative, which we can write as $-\gamma(5, y)$. Thus:

$$y^2 v = -\gamma(5, y) + C_1 \implies x' = v = -\frac{1}{y^2} \gamma(5, y) + \frac{C_1}{y^2}$$

To find $x(y)$, we integrate x' with respect to y :

$$x(y) = -\int \frac{1}{y^2} \gamma(5, y) dy - \frac{C_1}{y} + c_2$$

We solve the remaining integral using integration by parts. Let $u = \gamma(5, y)$ and $dw = -\frac{1}{y^2} dy$. Then $du = y^4 e^{-y} dy$ and $w = \frac{1}{y}$.

$$-\int \frac{1}{y^2} \gamma(5, y) dy = \frac{1}{y} \gamma(5, y) - \int \frac{1}{y} (y^4 e^{-y}) dy = \frac{1}{y} \gamma(5, y) - \int y^3 e^{-y} dy$$

The remaining integral is simply $\gamma(4, y)$. Substituting this back:

$$x(y) = \frac{1}{y} \gamma(5, y) - \gamma(4, y) - \frac{C_1}{y} + c_2$$

Factoring the $\frac{1}{y}$ terms to match the given solution form:

$$x(y) = \frac{1}{y} (\gamma(5, y) - C_1) - \gamma(4, y) + c_2$$

By setting $c_1 = -C_1$, we match the desired form exactly: $x(y) = \frac{1}{y} (\gamma(A, y) + c_1) - \gamma(B, y) + c_2$. Matching the arguments of the gamma functions, we easily identify $A = 5$ and $B = 4$. Therefore, $A + B = 9$.

Derivatives

3.1 Medium Derivative

Exponential Tangent Inverse Derivative

Let $f(x) = (x^2 + 1)^{\tan^{-1}(x)}$. Find $f'(1)$.

Solution: Because the function has x in both the base and the exponent, we utilize logarithmic differentiation. First, take the natural logarithm of both sides:

$$\ln(f(x)) = \ln\left((x^2 + 1)^{\tan^{-1}(x)}\right)$$

Using the power rule for logarithms, bring the exponent to the front:

$$\ln(f(x)) = \tan^{-1}(x) \ln(x^2 + 1)$$

Now, differentiate both sides implicitly with respect to x , applying the product rule on the right-hand side:

$$\frac{1}{f(x)} f'(x) = \frac{d}{dx}[\tan^{-1}(x)] \cdot \ln(x^2 + 1) + \tan^{-1}(x) \cdot \frac{d}{dx}[\ln(x^2 + 1)]$$

$$\frac{f'(x)}{f(x)} = \frac{1}{x^2 + 1} \ln(x^2 + 1) + \tan^{-1}(x) \left(\frac{2x}{x^2 + 1} \right)$$

Multiply both sides by $f(x)$ to isolate $f'(x)$:

$$f'(x) = (x^2 + 1)^{\tan^{-1}(x)} \left[\frac{\ln(x^2 + 1) + 2x \tan^{-1}(x)}{x^2 + 1} \right]$$

We now evaluate this derivative at $x = 1$. First, let's find the components:

$$\tan^{-1}(1) = \frac{\pi}{4}$$

$$\ln(1^2 + 1) = \ln(2)$$

$$f(1) = (1^2 + 1)^{\pi/4} = 2^{\pi/4}$$

Substituting these values directly into our $f'(x)$ expression:

$$f'(1) = 2^{\pi/4} \left[\frac{\ln(2) + 2(1)\left(\frac{\pi}{4}\right)}{1^2 + 1} \right]$$

$$f'(1) = 2^{\pi/4} \left[\frac{\ln(2) + \frac{\pi}{2}}{2} \right]$$

$$f'(1) = 2^{\pi/4} \left(\frac{2 \ln(2) + \pi}{4} \right)$$

3.2 Hard Derivative

Infinite Telescoping Series Derivative

Let $f(x) = \sum_{n=1}^{\infty} \tan^{-1} \left(\frac{\sqrt{x\sqrt{x\ldots}}}{1+nx^2+n^2x^2} \right)$. Find $f'(7)$.

Solution: First, we must simplify the infinitely nested radical in the numerator. Let $y = \sqrt{x\sqrt{x\ldots}}$. Squaring both sides gives $y^2 = x\sqrt{x\ldots}$, which implies $y^2 = xy$. Assuming $x > 0$, this yields $y = x$. Next, we factor the denominator of the argument:

$$1 + nx^2 + n^2x^2 = 1 + n(n+1)x^2 = 1 + (nx)((n+1)x)$$

The function simplifies to:

$$f(x) = \sum_{n=1}^{\infty} \tan^{-1} \left(\frac{x}{1 + (nx)((n+1)x)} \right)$$

Notice that the numerator x can be cleverly rewritten as the difference $(n+1)x - nx$. This transforms the argument into the structure of the arctangent difference identity: $\tan^{-1} \left(\frac{a-b}{1+ab} \right) = \tan^{-1}(a) - \tan^{-1}(b)$.

$$f(x) = \sum_{n=1}^{\infty} \tan^{-1} \left(\frac{(n+1)x - nx}{1 + (nx)((n+1)x)} \right) = \sum_{n=1}^{\infty} \left[\tan^{-1}((n+1)x) - \tan^{-1}(nx) \right]$$

This forms a telescoping series. Expanding the first few partial sums S_N :

$$S_N = (\tan^{-1}(2x) - \tan^{-1}(x)) + (\tan^{-1}(3x) - \tan^{-1}(2x)) + \cdots + (\tan^{-1}((N+1)x) - \tan^{-1}(Nx))$$

All intermediate terms cancel out, leaving:

$$S_N = \tan^{-1}((N+1)x) - \tan^{-1}(x)$$

Taking the limit as $N \rightarrow \infty$ (for $x > 0$), the first term approaches $\frac{\pi}{2}$:

$$f(x) = \lim_{N \rightarrow \infty} S_N = \frac{\pi}{2} - \tan^{-1}(x) = \cot^{-1}(x)$$

Now, we differentiate the simplified closed-form function $f(x)$:

$$f'(x) = -\frac{1}{1+x^2}$$

Evaluating this derivative exactly at $x = 7$:

$$f'(7) = -\frac{1}{1+7^2} = -\frac{1}{1+49} = -\frac{1}{50}$$

Physics & Engineering

4.1 Power Systems

Synchronous Generator Excitation

A 3-phase, 50 Hz, 11 kV synchronous generator has a synchronous reactance of 1.5Ω per phase and negligible armature resistance. It is delivering 10 MW at a power factor of 0.8 lagging to a large power grid maintained at 11 kV. Calculate the excitation voltage per phase in kV.

Solution: We analyze the system on a per-phase basis. The grid voltage acts as our reference phasor. The line-to-line grid voltage is 11 kV, so the per-phase terminal voltage V is:

$$V = \frac{11,000}{\sqrt{3}} \approx 6350.85 \angle 0^\circ \text{ V}$$

The total three-phase real power delivered is $P = 10 \text{ MW}$. We can find the magnitude of the line current I using the 3-phase power formula $P = \sqrt{3} V_L I_L \cos(\phi)$:

$$I = \frac{P}{\sqrt{3} V_L \cos(\phi)} = \frac{10 \times 10^6}{\sqrt{3} \times 11,000 \times 0.8} = \frac{10,000}{15.242} \approx 656.05 \text{ A}$$

The power factor is 0.8 lagging, meaning the current lags the voltage by an angle $\phi = \cos^{-1}(0.8) \approx 36.87^\circ$. Expressing the current as a complex phasor:

$$I = 656.05(\cos(-36.87^\circ) + j \sin(-36.87^\circ)) = 656.05(0.8 - j0.6) = 524.84 - j393.63 \text{ A}$$

The per-phase excitation (internal generated) voltage E is found using the synchronous machine voltage equation:

$$E = V + I(R_a + jX_s)$$

Since the armature resistance R_a is negligible, $R_a \approx 0$, and the synchronous reactance $X_s = 1.5 \Omega$:

$$E = 6350.85 + (524.84 - j393.63)(j1.5)$$

$$E = 6350.85 + j(787.26) - j^2(590.445)$$

Since $j^2 = -1$:

$$E = 6350.85 + 590.445 + j787.26 = 6941.30 + j787.26 \text{ V}$$

To find the required excitation voltage, we calculate the magnitude of this phasor:

$$|E| = \sqrt{6941.30^2 + 787.26^2} = \sqrt{48181645.69 + 619778.31} = \sqrt{48801424} \approx 6985.8 \text{ V}$$

Converting this per-phase voltage to kilovolts (kV):

$$|E| = 6.99 \text{ kV}$$

4.2 Quantum Mechanics

Infinite Potential Well Energy Transition

An electron is confined in a one-dimensional infinite potential well of width $L = 0.25$ nm. Determine the energy difference (in eV) between the second excited state and the ground state. Use: $m_e = 9.11 \times 10^{-31}$ kg, $h = 6.626 \times 10^{-34}$ J · s, and $1 \text{ eV} = 1.602 \times 10^{-19}$ J.

Solution: The energy levels for a particle in a 1D infinite potential well are quantized and given by the formula:

$$E_n = \frac{n^2 h^2}{8m_e L^2}$$

where n is the principal quantum number. The ground state corresponds to $n = 1$. The first excited state is $n = 2$, meaning the second excited state corresponds to $n = 3$. The energy of the ground state is $E_1 = \frac{1^2 h^2}{8m_e L^2}$. The energy of the second excited state is $E_3 = \frac{3^2 h^2}{8m_e L^2} = \frac{9h^2}{8m_e L^2}$. The energy difference ΔE is simply:

$$\Delta E = E_3 - E_1 = \frac{9h^2}{8m_e L^2} - \frac{h^2}{8m_e L^2} = \frac{8h^2}{8m_e L^2} = \frac{h^2}{m_e L^2}$$

We now plug in the physical constants to calculate this value in Joules. The width L must be converted to meters: $L = 0.25 \times 10^{-9}$ m.

$$\Delta E = \frac{(6.626 \times 10^{-34})^2}{(9.11 \times 10^{-31}) \times (0.25 \times 10^{-9})^2}$$

$$\Delta E = \frac{43.903876 \times 10^{-68}}{(9.11 \times 10^{-31}) \times (0.0625 \times 10^{-18})}$$

$$\Delta E = \frac{43.903876 \times 10^{-68}}{0.569375 \times 10^{-49}} \approx 77.1089 \times 10^{-19} \text{ J}$$

Finally, we convert the energy from Joules to electron-volts (eV) using the given conversion factor:

$$\Delta E \text{ (in eV)} = \frac{77.1089 \times 10^{-19}}{1.602 \times 10^{-19}} \approx 48.13 \text{ eV}$$

4.3 Mechanics

Gravitational Three-Body Linear System

A large spherical mass M is fixed at one position and two identical point masses m are kept on a line passing through the centre of M . The point masses are connected by a rigid massless rod of length l and this assembly is free to move along the line connecting them. All three masses interact only through their mutual gravitational interaction. When the point mass nearer to M is at a distance $r = 3l$ from M , the tension in the rod is zero for $m = k \left(\frac{M}{288} \right)$. Find the value of k .

Solution: Let the two identical point masses be m_1 and m_2 , where m_1 is closer to M . Their positions relative to the center of mass M are: - Distance of m_1 from M : $r_1 = 3l$ - Distance of m_2 from M : $r_2 = 3l + l = 4l$ Because the tension in the rigid rod connecting m_1 and m_2 is zero, the two masses must be experiencing the exact same acceleration strictly due to the gravitational forces acting upon them individually. Let's define the forces acting on each mass, taking the direction towards M as positive: For mass m_1 , the forces are the gravitational pull from M (positive direction) and the gravitational pull from m_2 (negative direction). Using Newton's Second Law:

$$ma_1 = \frac{GMm}{(3l)^2} - \frac{Gmm}{l^2} \Rightarrow a_1 = \frac{GM}{9l^2} - \frac{Gm}{l^2}$$

For mass m_2 , both the gravitational pull from M and the gravitational pull from m_1 act in the positive direction (towards M):

$$ma_2 = \frac{GMm}{(4l)^2} + \frac{Gmm}{l^2} \Rightarrow a_2 = \frac{GM}{16l^2} + \frac{Gm}{l^2}$$

Since the tension is zero, $a_1 = a_2 = a$. We equate the two acceleration expressions:

$$\frac{GM}{9l^2} - \frac{Gm}{l^2} = \frac{GM}{16l^2} + \frac{Gm}{l^2}$$

The l^2 terms cancel out entirely from the denominators:

$$\frac{GM}{9} - Gm = \frac{GM}{16} + Gm$$

Grouping the M terms on one side and the m terms on the other:

$$2Gm = \frac{GM}{9} - \frac{GM}{16} = GM \left(\frac{16-9}{144} \right) = GM \left(\frac{7}{144} \right)$$

Dividing both sides by $2G$ to solve for m :

$$m = \frac{7}{288}M$$

We are given that $m = k \left(\frac{M}{288} \right)$. By direct comparison, the constant k is:

$$k = 7$$